

Driving Efficiency Through Innovation Across the Operating Range and Power Chain

Aligning Power Architecture With the Needs of AI

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Oct 2026

Global Operations Enable Scale and Proximity

Company overview

$500M+
Annualized Run-Rate Revenue (USD)

1,500+
Employees

Global workforce including manufacturing

220+
Global R&D Staff

3
Manufacturing Sites

+26 Sites
Leveraging Kinpo/Compal Group Facilities

Global Headquarters

Plano, TX



Manufacturing Facilities

Matamoros, MX



Shanghai, CN



Thailand



Research & Development

Plano, TX



Shanghai, CN



North American Sales & Application Engineering

Milpitas, CA



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Four Pillars Turn Expertise into Deployment Confidence

Vertical Integration

- VAC-to-chip solutions · End-to-end integration · Supply-chain resilience
- Single-touch supplier

Proven Reliability

- Mission-critical heritage · Thermal expertise
- High MTBF & Low FIT metrics

Global Manufacturing

- Multi-site production · High-volume capacity · Regional supply proximity

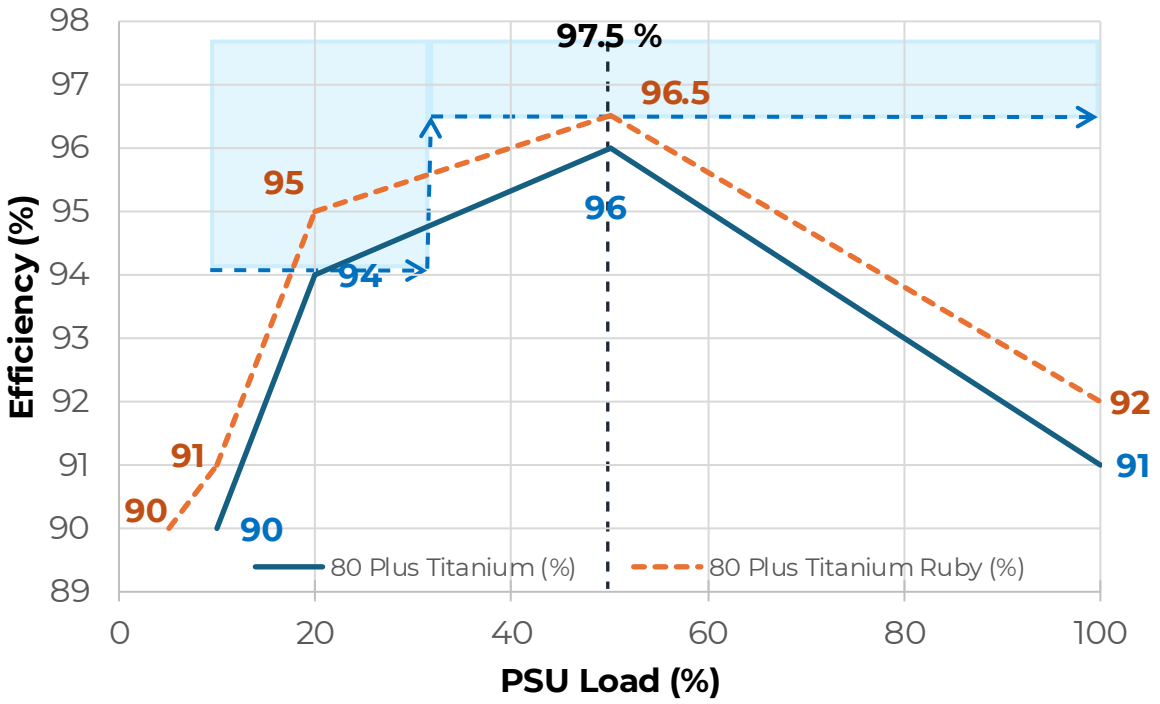
Design Intelligence

- System-level design · High-density AI infrastructure · Standards collaboration

AI Workloads Shift the Target to Operational Efficiency

Traditional server PSU selection (Front end)

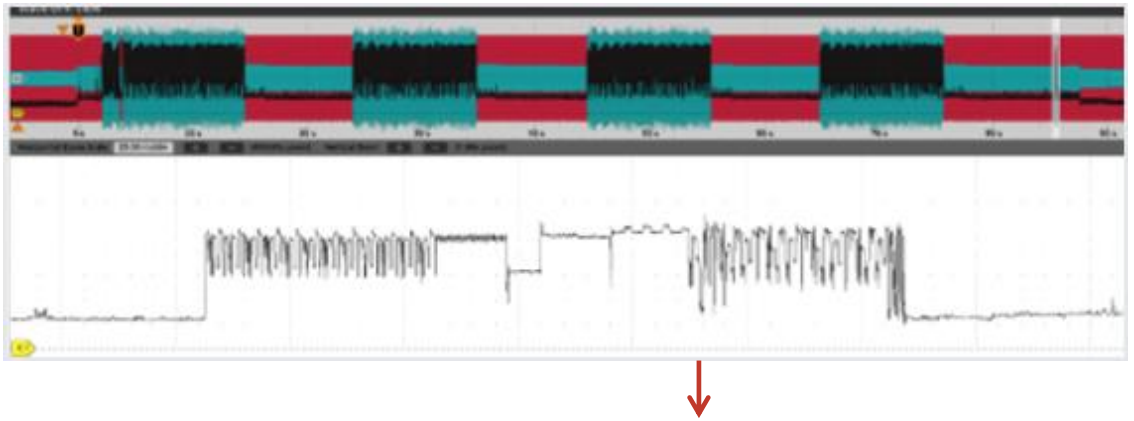
Peak efficiency at ~50% load was a primary benchmark



80 PLUS Titanium: 96% @ 50% load vs. 94% @ 20% load / 91% @ 100% load

AI/GPU workload-driven power profile

Optimize efficiency across the operating range, not at one point



Current swings within a few AC cycles

Millisecond zoom of instantaneous GPU current swings.

Real workload: continuous swings, no steady 50% point

Performance is set by time-weighted efficiency—not one peak

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Sources: 80 PLUS Titanium spec (230V redundant); Intel server PSU efficiency guidance (10/20/50/100% load).

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Technology Choice Follows the Operating Conditions

Si, SiC, and GaN offer different strengths; application requirements determine the optimum technology.

Mature

Silicon

$R_{DS(on)}$, typ.	**24 mΩ**
Gate charge (Qg)	**145 nC**
Reverse (VSD)	**1.0 V @ 35 A**
Recovery behavior	**Very high**
Switching capability	**Medium**

→

Established at High Power

Silicon carbide

$R_{DS(on)}$, typ.	**20 mΩ**
Gate charge (Qg)	**67 nC**
Reverse (VSD)	**3.9 V @ 32 A**
Recovery behavior	**Higher**
Switching capability	**High**

→

Rapidly Advancing

Gallium nitride

$R_{DS(on)}$, typ.	**23 mΩ**
Gate charge (Qg)	**45 nC**
Reverse (VSD)	**1.0 V @ 43 A**
Recovery behavior	**Lower**
Switching capability	**Very high**

Totem-Pole PFC
(Hard Switching)

↓ Q_g + ↓ E_{oss} + ↓ V_{SD} + ↓ Qrr
lower switching/commutation loss

LLC
(Soft Switching)

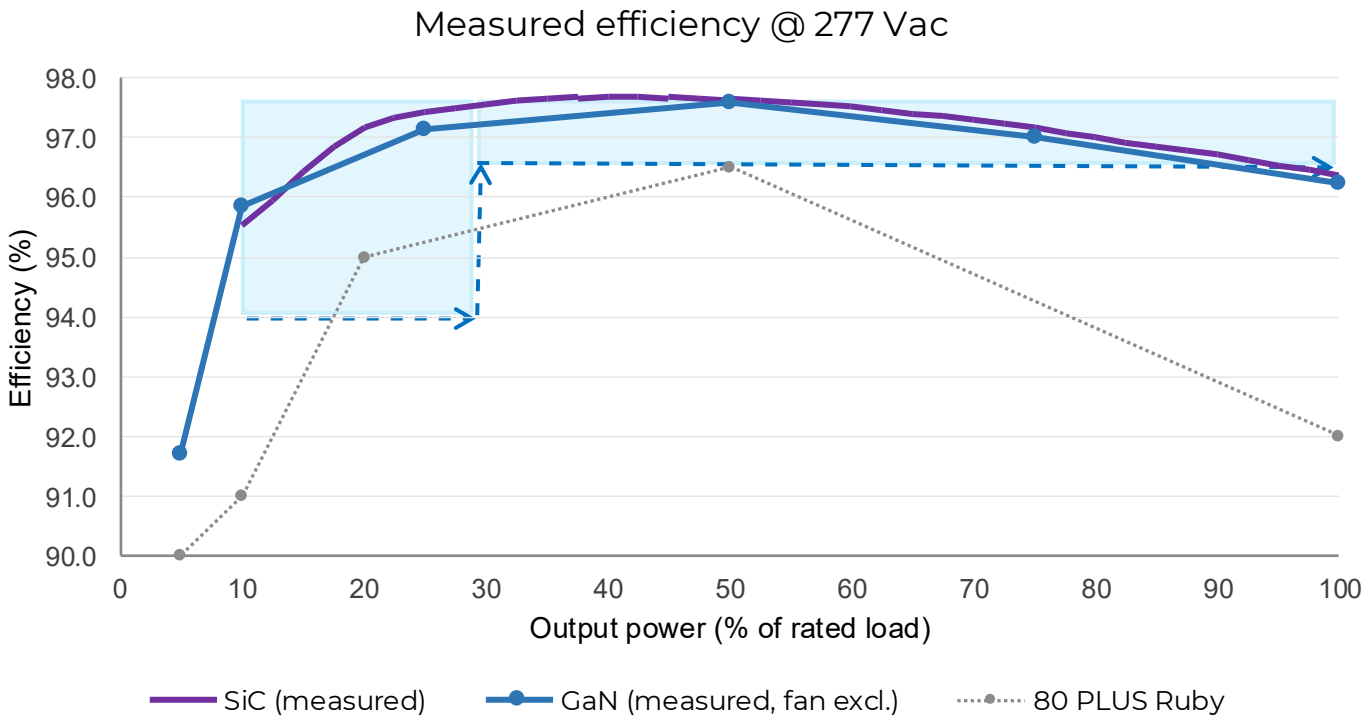
↓ Q_g + ↓ Q_{oss} + ↓ V_{SD}
lower dead time + higher frequency

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Two Technology Paths, Both High Performance

Technology choice follows the load profile, cost target, and maturity



- **Both paths reach ~97.6–97.7% peak efficiency at 277 Vac**
- At every measured point from 10% to 100% load, the gap is ≤0.3 pts
- Remaining differences track load point, control strategy (e.g., phase shedding) and test conditions
- **Developing both creates optionality: select by load profile, cost, and supply needs**

Load	SiC	GaN[1]	Δ (pts)
10%	95.54%	95.85%	0.31
50%	97.65%	97.60%	0.05
100%	96.38%	96.23%	0.15

Cost
- GaN: cost potential via standard Si fab equipment
- SiC: cost reducing as volumes scale

Maturity
- GaN: scaling volumes; 12 kW designs demonstrated
- SiC: multiple qualified sources

Power Density
- GaN: higher frequency, smaller magnetics
- SiC: proven high-power thermals

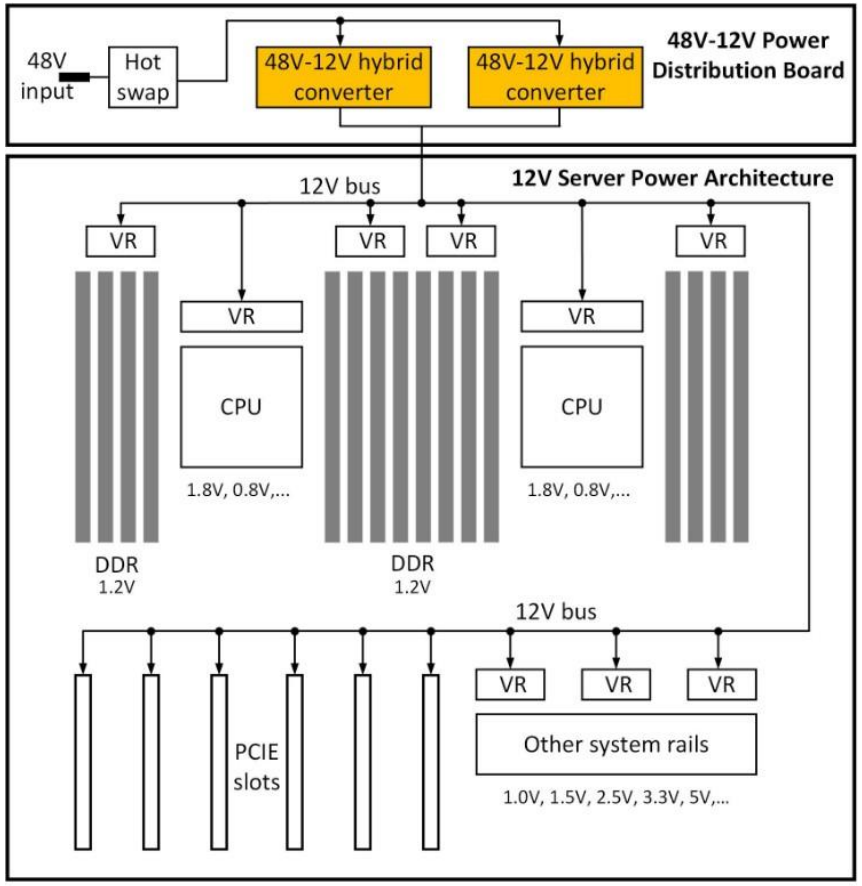
Test conditions differ; comparison is indicative, not a like-for-like benchmark. SiC: OmniOn measured data, 277 Vac, fan power included (curve digitized from test plot). GaN: third-party 12 kW reference design test report (Jun 2026), 277 Vac, phase shedding enabled, fan power excluded (fan est. ~33 W at 12 kW ≈ 0.27 pts); plotted at measured points only. [1]Fan exclusion favors GaN. Reference: 80 PLUS Ruby minimum efficiency (5/10/20/50/100% load).

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48 V Architectures Bridge Rack Power to GPU Loads

Li and S. Jiang, “Google 48V Rack Adaptation and Onboard Power Technology Update,” 2019



OmniOn POWER
Osprey
1/4 Brick

ARCHITECTURE IMPLICATION

Regulated 50 V-to-12 V conversion enables quick adaptation of 12 V-based workloads to 50 V systems.

DESIGN TARGET

Sustain high efficiency across every conversion stage as the operational load moves.

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GPU Load Swings Make Flat Efficiency the Design Target

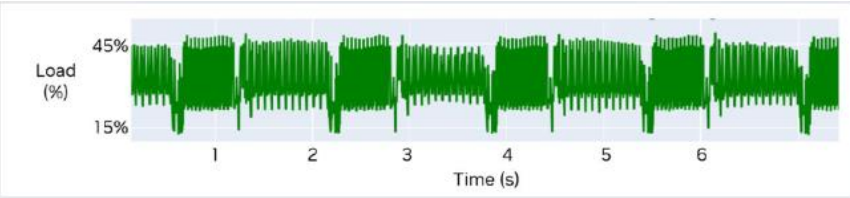
Utilization variability broadens the downstream converter's operating range

01
GPU demand varies
Utilization swings ~15–45% during LLM workloads.

02
Power range widens
50 V→12 V conversion operates ~45–75%, with higher peaks.

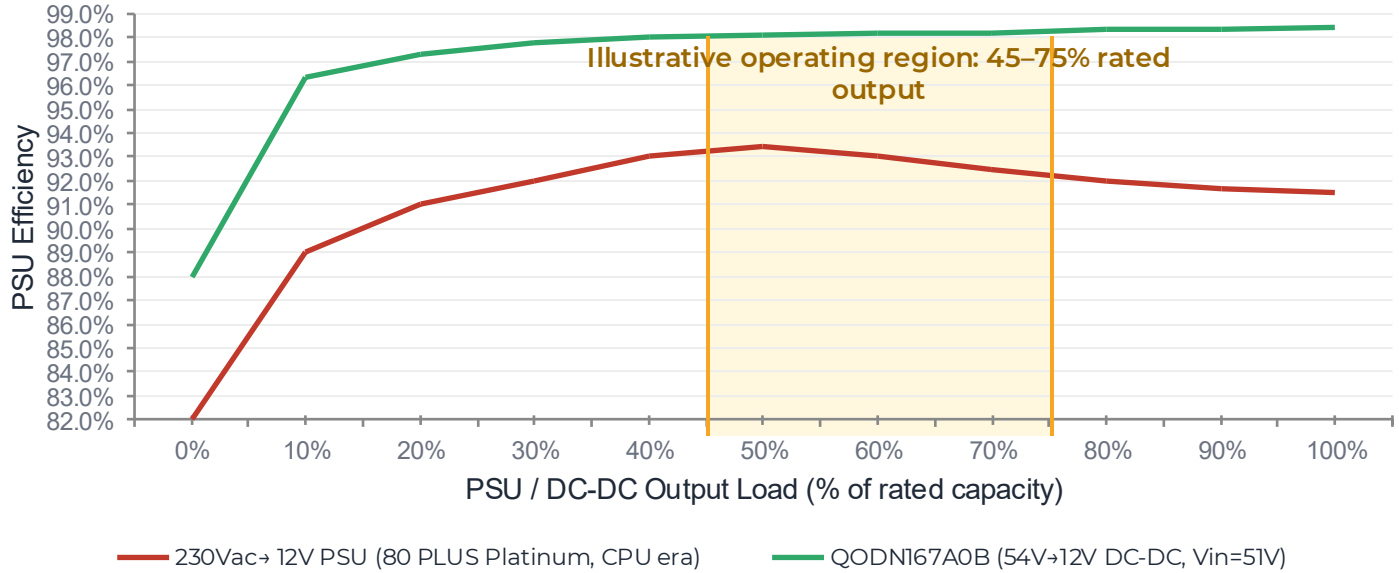
03
Flat efficiency wins
Sustain >98% from 30–100% load with <0.5-point variation.

WORKLOAD EVIDENCE



GPU utilization repeatedly swings between ~15% and ~45%.

POWER-CONVERSION EVIDENCE



Measured 54 V→12 V DC/DC stays >98% across 30–100% load.

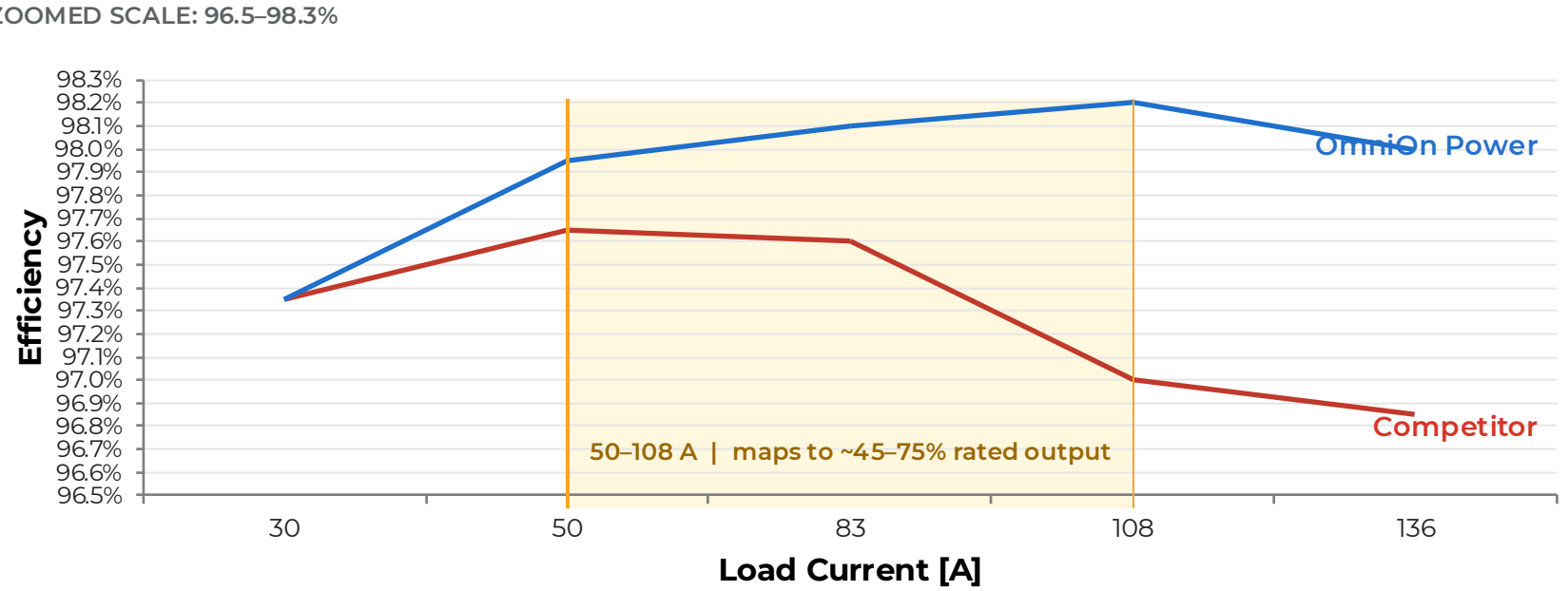
Sources: NVIDIA LLM workload trace; OmniOn measured QODN167A0B DC/DC data; 80 PLUS Platinum PSU reference curve.

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OmniOn Holds Efficiency Within 0.23 Points Across the GPU Operating Range

0.23-point variation versus 0.61 points from 50–108 A



MEASURED RESULT

0.23%

OmniOn variation

0.61%

Competitor variation

Flatter loss profile → more predictable heat output.

Derived from efficiency data; not a direct thermal measurement.

ILLUSTRATIVE VALUE SCENARIO

$1.8M projected 5-year NPV at 8%

1,000 racks · 132 kW/rack

Continuous operation · 8,760 hrs/yr

PUE 1.3 · $0.05/kWh

0.59-point current-weighted efficiency gain

Measured bench result shown above; NPV is an illustrative scenario under the stated assumptions. Full method is in speaker notes.

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Osprey Regulates 50 V to 12 V for High-Power GPUs

Power levels: 1.3, 1.6, 2.0, 2.6, and 3.5 kW · Non-Isolated Bus Converter · GPU-Ready

50 : 12 V_{DC}
Nom. Voltage Ratio
12V_{DC} regulated

2.6 kW
Max Output Power
2.6 kW variant

0.30%
Mid-range Efficiency
25% to 75% load

INPUT, POWER, & OUTPUT

Input Voltage and Output Voltage
40 – 60 V_{DC} : 12 V_{DC} (regulated)

Efficiency
Load: 50% and 100%: 98.5 and 97.9%

Peak Power
4.0 kW @ 3ms

INTERFACE & CONTROL

Communications
PMBus configuration

Firmware Update / Remote Control
Over I^2C / On/Off + Remote Sense

MECHANICAL, PROTECTION, THERMAL, & MANUFACTURING

Protection
Input: OVP, Output: OCP / OTP / SCP

Operating Temperature (wide)
-40°C to 85°C

Height Variants and Mounting
13.6 mm / 14.5 mm / custom (1/4 brick)
Heat plate + heatsink holes

Parallel Operation
Yes – trace routing and cooling important

Sustainability
Halogen free solutions available

Manufacturing
Two source manufacturing

Form Factor: ¼ Brick



Osprey BMP Module · 13.6 / 14.5 mm height variants

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Osprey DCDC Portfolio Scales the Efficiency Advantage

Module	Power Rating (W)	Input Range (Vin)	Output Current (A)	Peak Efficiency (50% load)	Time-weighted Efficiency (25% to 75%)	Full Load Efficiency (%)	Peak Power 3ms (W)
QODN108A0B	1300	40–60 V	108	98.5%	0.83%	98.1	1800
QODN136A0B	1600	40–60 V	136	98.5%	0.48%	98.1	2400
QODN167A0B	2000	40–60 V	167	98.5%	0.60%	98.1	3000
QODN217A0B	2600	40–60 V	217	98.5%	0.30%	97.9	4000
QODN292A0B	3500	40–60 V	292	In Development			

Continuous innovation extends power and efficiency across the portfolio.

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Innovation Delivers Efficiency Across the Full Power Chain

Scale, workload-led design, and measured performance create an AI infrastructure advantage

01

Global Scale You Can Rely On

- Global manufacturing, and R&D provide scale and proximity
- Vertical integration spans end-to-end through a single supplier
- Reliability, thermal expertise, and design intelligence support mission-critical deployment

02

Designed for Real AI Workloads

- Dynamic AI loads make time-weighted efficiency, not one peak, the real target
- Silicon, SiC, and GaN create options matched to load, cost, and maturity
- 48 V architectures require stage-by-stage optimization from rack input to PoL

03

Proven Performance Across the Range

- SiC and GaN paths each reach >97.5% peak efficiency at 277 Vac
- Bench data shows improved flatness with substantial OPEX savings
- Osprey scales regulated 50 V-to-12 V conversion from 1.3 to 3.5 kW

Bottom line | Follow the workload. Optimize the full power chain and not one peak.

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